



Army-615: Open-Source Threat Pattern Detection for Force Protection

SPONSORING ORGANIZATION

United States Army

CHALLENGE

Garrison and senior installation leaders cannot systematically monitor, correlate, and assess open-source online discussions related to personnel movements, high-visibility events, and senior leader presence, resulting in missed early warning indicators and increased risk to force protection, senior leaders, their families, and critical events.

RELEVANT CONTEXT

- The modern information environment is shaped by pervasive, user-generated content across open-source platforms such as Reddit, online forums, social media networks, niche community boards, and even the dark web where civilians routinely document and discuss unusual aircraft, convoys, security postures, and public events involving military or government presence.
- Observations shared online are often benign in isolation, but when aggregated across platforms, time, and geography, they can reveal sensitive patterns related to movements, intent, and protection postures of high-visibility events such as presidential visits, senior leader travel, military participation in public venues, or large community events.
- Civilians increasingly possess sophisticated tools such as high-resolution cameras, flight tracking applications, geolocation services, and open mapping platforms that amplify the fidelity and speed at which sensitive observations enter the public domain.
- Online communities frequently collaborate to answer questions, validate sightings, and fill information gaps, unintentionally accelerating the synthesis of sensitive information without any single authoritative source. For example, when certain rare, uniquely configured or geographically constrained military units, platforms, or aircraft make public appearances, informed observers can make inferences of VIP presence or mission type based on limited visual or auditory cues.
- Intelligence processes emphasize structured analytic tradecraft and formal reporting standards, which are often optimized for classified inputs and deliberate analytic timelines rather than rapid, ambiguous open-source signal detection.
- Currently, there is no unified, real-time capability to ingest, monitor, and analyze multiple open, semi-open, and illicit online spaces simultaneously for discussions tied to specific locations, individuals, or events. Existing tools are limited in their ability to correlate usernames, posting behavior, linguistic patterns, and thematic overlaps across platforms, including migration between mainstream forums and the dark web. Analysts and protection teams lack integrated dashboards that surface spikes in discussion volume, recurring usernames, geographic clustering, or converging lines of inquiry that may indicate emerging targeting behavior.
- Classification constraints, legal considerations, and the risk of confirmation bias can inhibit internal government entities from fully exploiting ambiguous open-source information without inadvertently signaling knowledge or validating sensitive facts.



- An independent or external analytic capability could observe, assess, and flag concerning open-source patterns without confirming or denying classified information, preserving operational security while enabling earlier warning.

IMPACT

If this problem were solved, it would provide garrison leaders, protection professionals, and senior commanders with earlier visibility into emerging threats within the open-source information environment, enabling proactive risk mitigation rather than reactive response and enhancing force protection by identifying potential targeting indicators before incidents occur, improving accountability by tracing how sensitive information spreads online, and reducing investigative friction by allowing law enforcement and security agencies to begin with contextualized insights rather than starting from zero.

POTENTIAL BENEFICIARIES

Garrison commanders and installation leadership; Protection warfighting function professionals; Force protection and physical security officers; Intelligence analysts supporting installation and domestic operations; Senior leaders and high-ranking officials; Military families and civilian personnel associated with installations

TEAM RECOMMENDED SKILL SETS

Computer Science and Software Engineering, Machine Learning and Artificial Intelligence, Data Engineering and Data Visualization, Cybersecurity

PROBLEM SPONSOR

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